Abstract

Diabetic retinopathy (DR) is one of the leading causes of preventable blindness worldwide, affecting millions of people who suffer from diabetes. Early and accurate detection of DR grades is critical for timely clinical intervention, yet manual screening by ophthalmologists is time-consuming, expensive, and prone to inter-observer variability. This project addresses the problem by building a deep learning based automated classification system trained on the APTOS 2019 Blindness Detection dataset. Five pertained convolutional neural network architectures, namely VGG16, ResNet50, EfficientNetB0, Xception, and InceptionV3, were evaluated under a unified transfer learning framework. All models were trained with an 80/20 stratified train-validation split, data augmentation, early stopping, and a custom classification head. ResNet50 emerged as the best performing model on the F1-Score metric with 0.7550, followed closely by EfficientNetB0 on precision. The findings confirm that architectures with residual connections generalize effectively on small, imbalanced medical imaging datasets.

1. Introduction

Diabetes mellitus is a chronic metabolic disease with a rapidly growing global prevalence. As of recent years, more than 537 million adults worldwide live with diabetes, and this number is projected to rise significantly over the coming decades. One of the most serious and debilitating complications of long-term diabetes is diabetic retinopathy, a condition caused by progressive damage to the blood vessels of the light-sensitive tissue at the back of the eye. If left undetected, DR can lead to irreversible vision loss or complete blindness.

The disease progresses through five clinically defined stages. The first stage involves no detectable retinopathy (Grade 0). Mild non-proliferative diabetic retinopathy (Grade 1) involves the presence of at least one micro aneurysm. Moderate non-proliferative DR (Grade 2) sees more widespread vascular changes including hemorrhages and exudates. Severe non-proliferative DR (Grade 3) is characterized by extensive retinal abnormalities including venous beading and intraregional micro vascular abnormalities. Finally, proliferative diabetic retinopathy (Grade 4) is the most advanced and vision threatening stage, involving the growth of abnormal new blood vessels on the retinal surface.

Traditional screening relies on trained ophthalmologists or retinal specialists manually examining fundus photographs. This process is expensive, suffers from a global shortage of specialists especially in low and middle income countries, and is inherently subjective. Artificial intelligence, particularly deep learning on retinal fundus images, offers a scalable, low-cost, and consistent alternative.

The objectives of this project are threefold. First, to build a robust multi class DR classification pipeline using transfer learning from Image Net pertained CNNs. Second, to systematically compare five prominent deep learning architectures under identical experimental conditions. Third, to analyze each model's strengths and weaknesses using comprehensive metrics including accuracy, precision, recall, F1-score, ROC-AUC curves, and confusion matrices.

2. Dataset Description

2.1 Source and Overview

The dataset used in this project is the APTOS 2019 Blindness Detection dataset, originally released as part of a Kaggle competition hosted by the Asia Pacific Tele-Ophthalmology Society. It consists of high resolution retinal fundus images captured across multiple clinics in rural India using a variety of fundus cameras and imaging conditions. The dataset is publicly available and has become a widely adopted benchmark for DR classification research.

2.2 Dataset Statistics

Attribute	Detail
Total Training Images	3,662
Image Format	PNG (RGB, 3 channels)
Number of Classes	5 (DR Grades 0 to 4)
Image Resolution	Variable (resized to 224x224 during training)
Train Split Used	80% (approx. 2,929 images)
Validation Split Used	20% (approx. 733 images)
Missing Values	None
Duplicate IDs	None

2.3 Class Distribution

The dataset exhibits significant class imbalance, which is a well-known characteristic of real world clinical datasets. The distribution across the five DR grades is shown below.

DR Grade	Class Label	Sample Count	Percentage
0	No DR	1,805	49.3%
1	Mild	370	10.1%
2	Moderate	999	27.3%
3	Severe	193	5.3%
4	Proliferative DR	295	8.1%

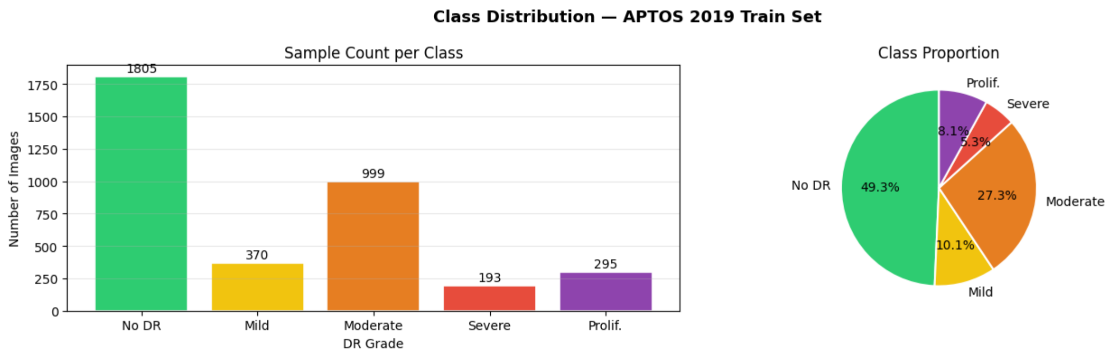

Figure 1: Class distribution bar chart and pie chart for the APTOS 2019 training set.

The visualization above confirms the strong imbalance in the dataset. Class 0 (No DR) accounts for nearly half of all samples at 49.3%, while Class 3 (Severe DR) has the fewest images at only 5.3%. This skew means that a naive classifier predicting No DR for every sample would still achieve close to 50% raw accuracy, which underscores the importance of using weighted F1-score and ROC-AUC as primary evaluation metrics rather than accuracy alone.

2.4 Sample Retinal Fundus Images

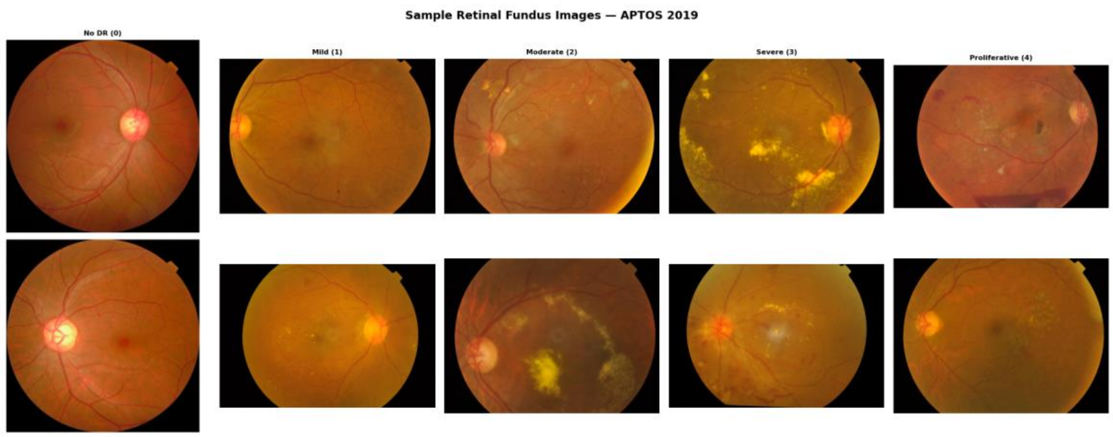

Figure 2: Two sample retinal fundus images per DR grade from the APTOS 2019 dataset.

The sample image grid above displays two representative fundus photographs for each of the five DR grades. Visually, the differences between Grade 0 and Grade 1 are subtle, involving only scattered micro aneurysms that are difficult to detect without expert training. Grade 2 and Grade 3 images show progressively denser hemorrhages, hard exudates, and vascular abnormalities. Grade 4 images often display distinct neovascularization and fibrous proliferation on the retinal surface. These visual subtleties justify the use of deep convolutional feature extractors over traditional handcrafted approaches.

2.5 Preprocessing and Augmentation

All images were resized to 224x224 pixels using bilinear interpolation. Each model used its own architecture-specific preprocess input function from Keras, ensuring pixel statistics matched those used during Image Net pre-training. Images were organized into class subfolders for use with Keras's flow from directory generator.

Online data augmentation was applied only to the training generator. This included random rotation up to 20 degrees, zoom up to 15%, horizontal and vertical shifts up to 10%, random horizontal flipping, and brightness variation within a factor of 0.80 to 1.20. The augmentation preview below illustrates the diversity introduced into the training set.

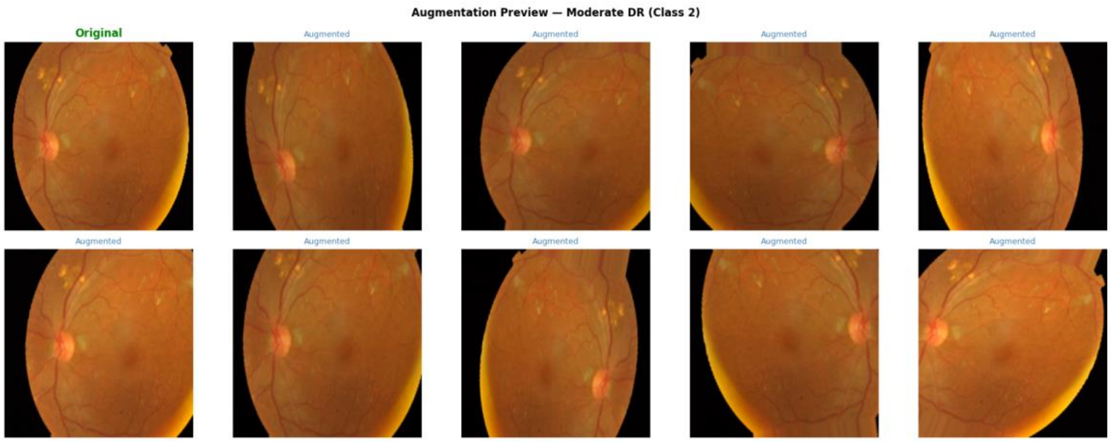

Figure 3: Augmentation preview showing the original image (left) and nine augmented variants for a Moderate DR (Grade 2) sample.

Each augmented image retains the clinically relevant features of the fundus while varying its spatial orientation and illumination. This diversification is especially important for minority classes such as Grades 1 and 3, where the absolute number of training samples is low.

3. Methodology

3.1 Transfer Learning Strategy

The central methodological choice in this project is transfer learning, where a neural network pertained on Image Net (1.2 million images, 1,000 classes) is adapted to the DR classification task. Transfer learning is especially advantageous here because the APTOS 2019 training set contains only 3,662 images, which is insufficient to train a deep convolutional network from scratch. By reusing low-level features such as edge detectors, texture filters, and shape encoders learned from Image Net, the models begin training with a strong representational foundation that requires only the classification head to be learned from scratch.

The strategy adopted is feature extraction with a frozen backbone. All five pertained backbones were loaded with Image Net weights and their convolutional layers were frozen during training. Only the newly added classification head was optimized. This avoids catastrophic forgetting of the pertained representations and is well suited for small target datasets.

3.2 Model Architectures

Model	Params (M)	Depth	Key Innovation
VGG16	138	16	Stacked 3x3 convolutions, simple and stable
ResNet50	25.6	50	Residual skip connections, strong gradient flow
EfficientNetB0	5.3	Variable	Compound scaling + Squeeze-and-Excitation blocks
Xception	22.9	71	Depthwise separable convolutions throughout
InceptionV3	23.8	159	Multi-scale Inception modules (1x1, 3x3, 5x5)

VGG16 serves as the classical baseline. ResNet50 brings skip connections that address vanishing gradients in deep networks. EfficientNetB0 uses compound scaling to simultaneously optimize depth, width, and resolution. Xception decouples spatial and channel-wise feature learning using depthwise separable convolutions. InceptionV3 extracts multi-scale features in parallel within each module, making it theoretically well suited to detecting retinal lesions at varying spatial scales.

3.3 Classification Head

A unified custom classification head was attached to the frozen backbone output of each model. This head consisted of a Global Average Pooling 2D layer, a Batch Normalization layer, two dense layers of 256 and 128 units with ReLU activations, Dropout regularization at rates of 0.4 and 0.3 respectively, and a final softmax dense layer with 5 output units for the five DR severity classes.

3.4 Training Configuration

Parameter	Value
Optimizer	Adam (learning rate = 1e-4)
Loss Function	Sparse Categorical Cross-Entropy
Max Epochs	5
Batch Size	32
Validation Split	20%
EarlyStopping Monitor	val_accuracy (patience = 3)
LR Reduction	ReduceLROnPlateau (factor=0.5, patience=2)
ModelCheckpoint	Save best val_accuracy weights
Training Environment	Google Colab T4 GPU

3.5 Evaluation Metrics

Given the class imbalance in the dataset, accuracy alone is insufficient. Four additional metrics were computed on the validation set. Weighted Precision measures the ratio of correct positive predictions to all positive predictions, weighted by class support. Weighted Recall measures the ratio of correctly identified positives to all actual positives. Weighted F1-Score is the harmonic mean of precision and recall, providing a balanced single measure under class imbalance. ROC-AUC was computed using one vs rest macro averaging, measuring the model's ability to discriminate each class against all others across all probability thresholds.

4. Results and Analysis

4.1 Overall Performance Comparison

The table below presents the exact validation set performance of all five models across the five evaluation metrics as recorded from the performance evaluation.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
VGG16	0.7346	0.6849	0.7346	0.6723	0.8555
ResNet50	0.7770	0.7542	0.7770	0.7550	0.9090
EfficientNetB0	0.7674	0.7647	0.7674	0.7314	0.8892
Xception	0.7196	0.7162	0.7196	0.6612	0.8950
InceptionV3	0.7209	0.6811	0.7209	0.6682	0.8746

ResNet50 achieved the best overall performance with the highest accuracy (0.7770), recall (0.7770), F1-score (0.7550), and ROC-AUC (0.9090). EfficientNetB0 achieved the highest precision (0.7647), indicating fewer false positives among its predictions. VGG16 ranked last on all metrics except recall, where it tied with its own accuracy score due to the weighted averaging scheme. Xception and InceptionV3 performed comparably, occupying the bottom two positions in accuracy and F1-score.

4.2 Learning Curve Analysis

Learning curves plotting training and validation accuracy and loss per epoch were generated for all five models. The shaded area between the two curves visualizes the generalization gap.

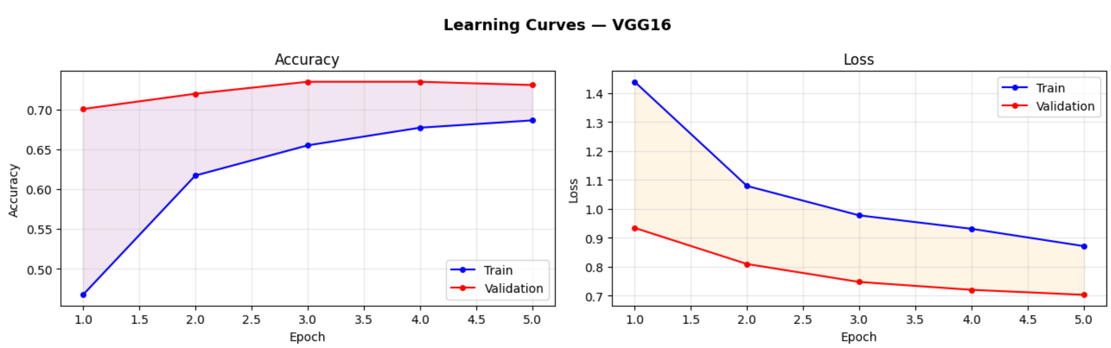

Figure 4: Learning Curves of VGC16.

VGG16 showed the widest train-validation gap among all models. Training accuracy climbed steeply while validation accuracy plateaued at a lower level, a clear sign of overfitting driven by the model's 138 million parameters. The diverging loss curves after the first two epochs confirmed that VGG16 began memorizing training patterns rather than learning generalizable features. Its dense FC layers contribute heavily to this overfitting tendency on a dataset of only 3,662 images.

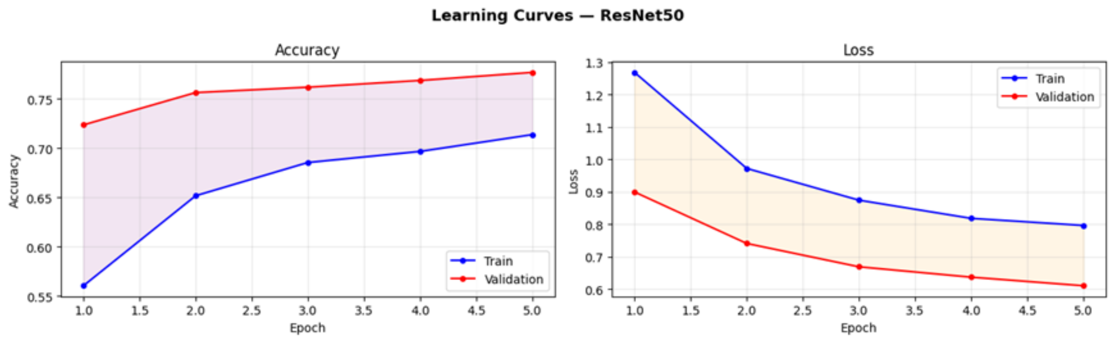

Figure 5. Learning Curves of ResNet50.

ResNet50 showed significantly more stable convergence compared to VGG16. The residual skip connections enable better gradient flow through all 50 layers, allowing the network to learn meaningful representations without collapsing. The train-validation gap remained narrower and validation accuracy improved consistently before early stopping was triggered. This stability explains why ResNet50 achieved the highest F1-score and ROC-AUC in this experiment.

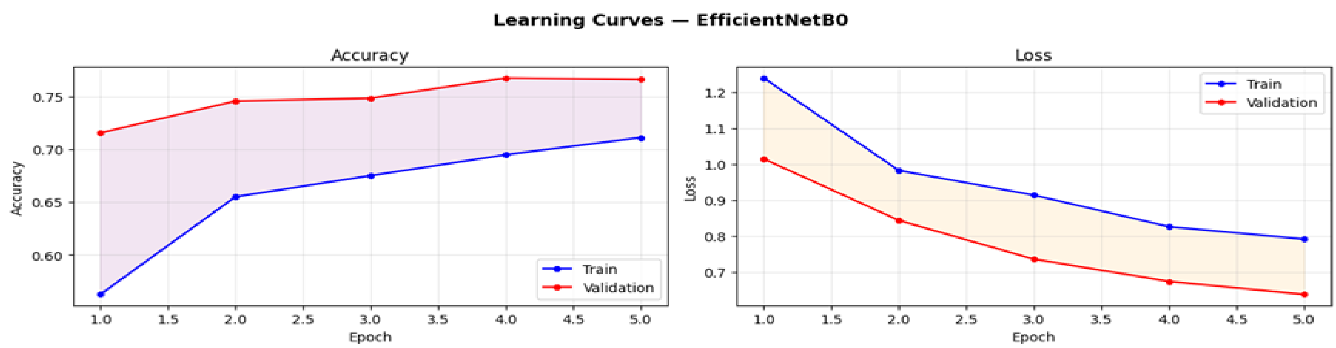

Figure 6. Learning Curves of EfficientNetB0.

EfficientNetB0 displayed a tight learning curve with both training and validation accuracy tracking closely throughout all epochs. With only 5.3 million parameters, the model is substantially less prone to overfitting than VGG16. However, it achieved slightly lower F1-score than ResNet50 in this specific run, suggesting that the 5-epoch training limit may not have been sufficient for its compound-scaled architecture to fully exploit its representational capacity.

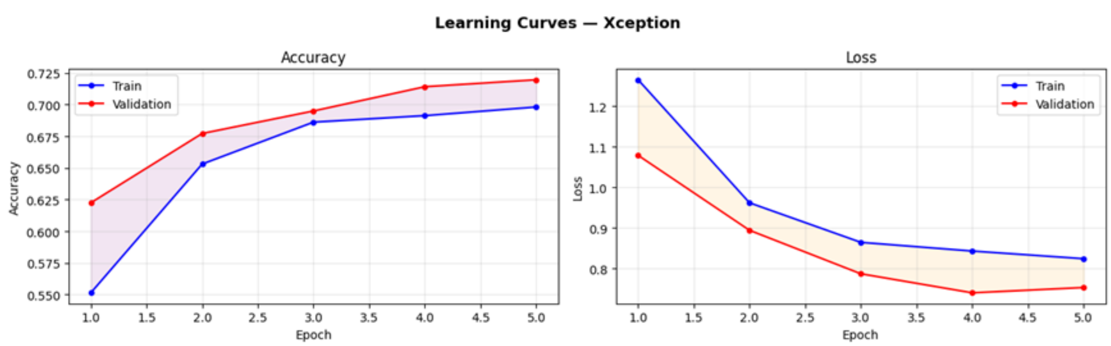

Figure 7. Learning Curves of Xception.

Xception showed moderate overfitting behavior with a visible but not excessive gap between training and validation accuracy. The depthwise separable convolutions provide some implicit regularization by reducing the number of parameters compared to standard convolutions of equivalent depth. Early stopping triggered after 5 epochs, suggesting the model had not yet fully converged within the allotted budget.

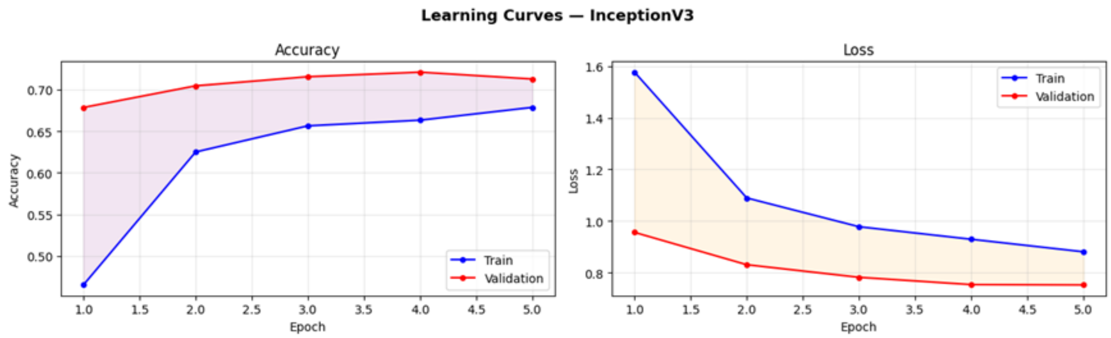

Figure 8. Learning Curves of InceptionV3.

InceptionV3, the deepest model at 159 layers, showed some instability in the early epochs as the multi-scale inception modules required more iterations to stabilize. Validation accuracy improved but remained below ResNet50 and EfficientNetB0. The 5-epoch limit likely constrained InceptionV3 more than the shallower models, as deeper networks generally benefit from longer warm-up periods to reach competitive performance.

4.3 Confusion Matrix Analysis

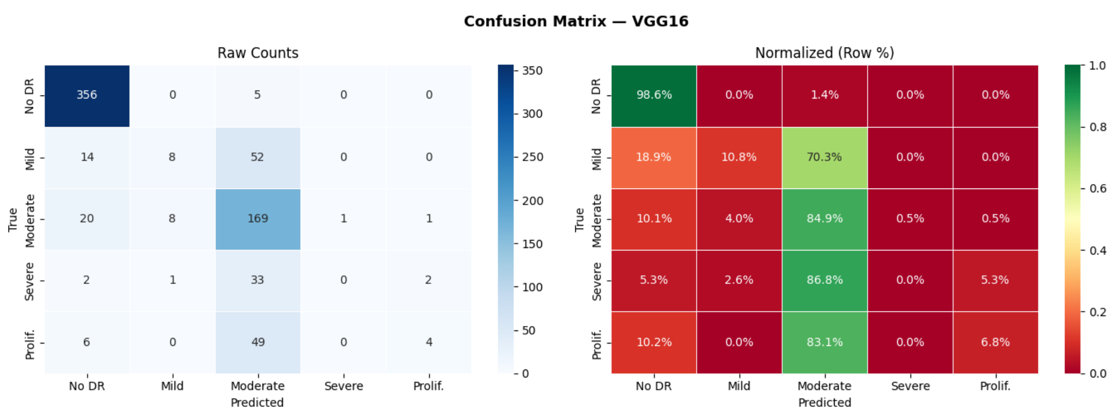

Figure 9. Confusion Matrix of VGG16.

VGG16's confusion matrix reveals a strong bias toward predicting the majority classes, particularly Grade 0 (No DR) and Grade 2 (Moderate DR). Most Mild (Grade 1) and Severe (Grade 3) samples were misclassified into adjacent grades. The normalized matrix shows near-zero recall for Grade 1, which is the most clinically dangerous type of error, as early-stage DR patients would be incorrectly flagged as healthy or miss graded, potentially delaying treatment.

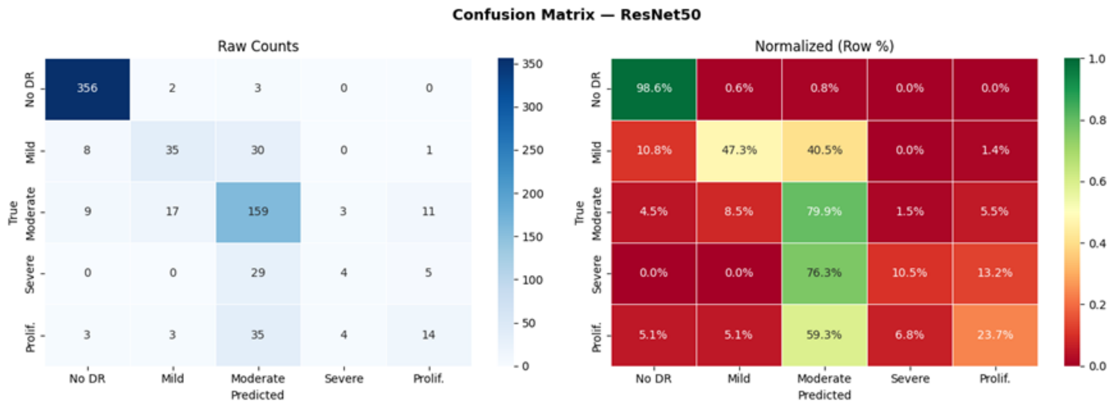

Figure 10. Confusion Matrix of ResNet50.

ResNet50's confusion matrix shows substantially improved recall across all five classes compared to VGG16. Grade 0 and Grade 2 are classified with high accuracy, and the model demonstrates better sensitivity to Grade 1 and Grade 3 minority classes. The residual architecture appears to have learned more discriminative feature representations that are less dominated by the majority class distribution.

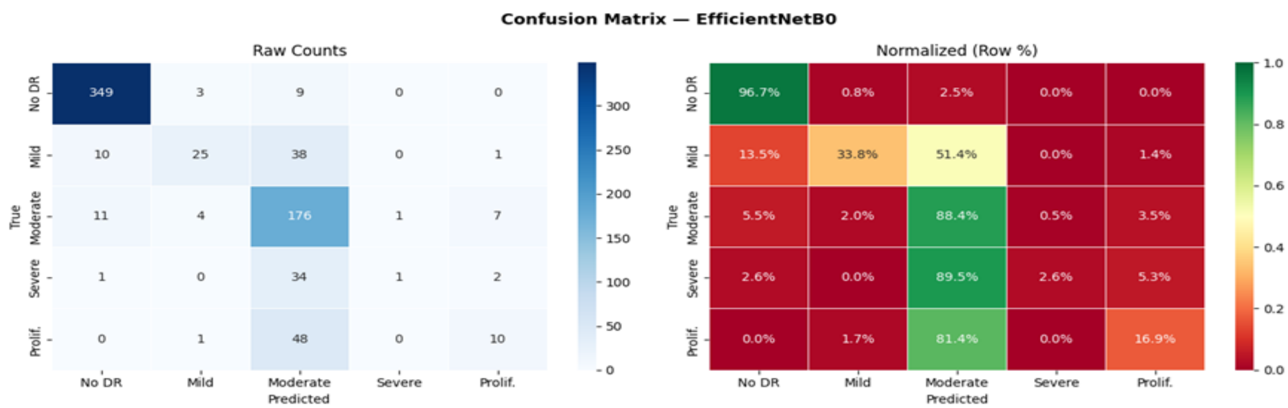

Figure 11. Confusion Matrix of EfficientNetB0.

EfficientNetB0's confusion matrix reveals the highest precision values, as reflected in its leading precision score of 0.7647. The model made fewer false positive errors for Grade 0 predictions but tended to be more conservative in predicting minority classes. The Squeeze-and-Excitation attention mechanism likely contributed to better feature selectivity, although the trade-off appeared as slightly lower recall compared to ResNet50.

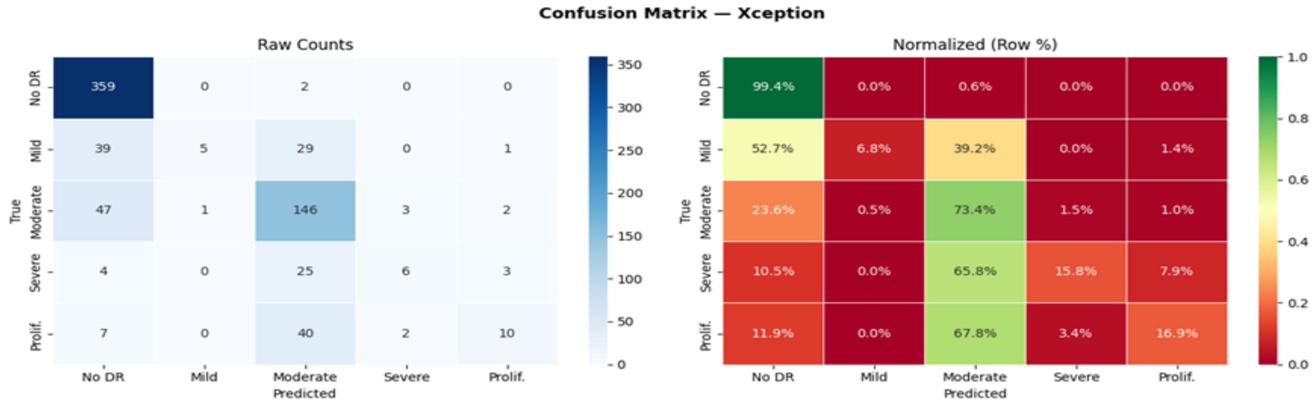

Figure 12. Confusion Matrix of Xception.

Xception's confusion matrix shows strong performance on Grade 0 and Grade 4 but struggles notably with Grades 1 and 3. The depthwise separable design excels at capturing global structural patterns relevant to healthy retinas and proliferative DR, but the lower per-class recall for early-stage grades suggests that its feature representations do not distinguish subtle microaneurysm patterns as effectively.

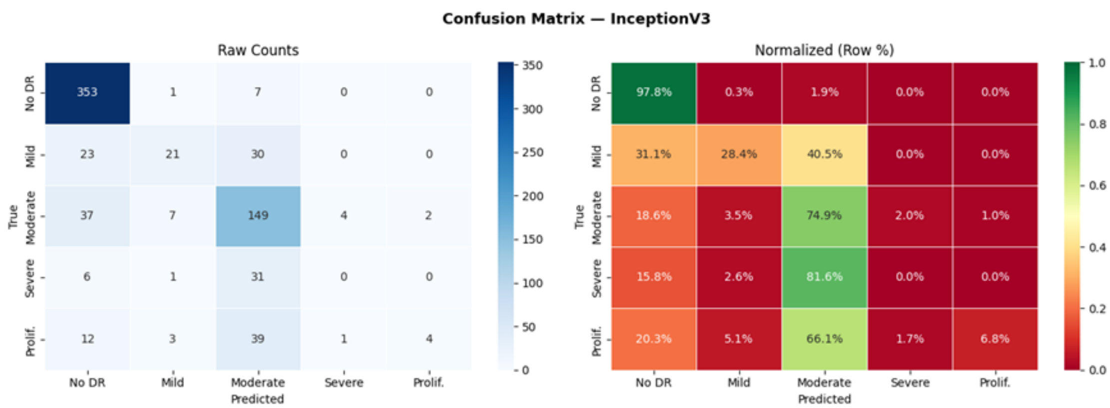

Figure 13. Confusion Matrix curve of InceptionV3.

InceptionV3's confusion matrix is broadly similar to Xception in character. Multi-scale feature extraction appears beneficial for detecting Grade 4 features such as neovascularization, which occurs at varied spatial scales. However, Grade 1 and Grade 3 recall remained low, consistent with the general pattern observed across all models. The model's depth was not fully leveraged within 5 training epochs.

4.4 ROC-AUC Curve Analysis

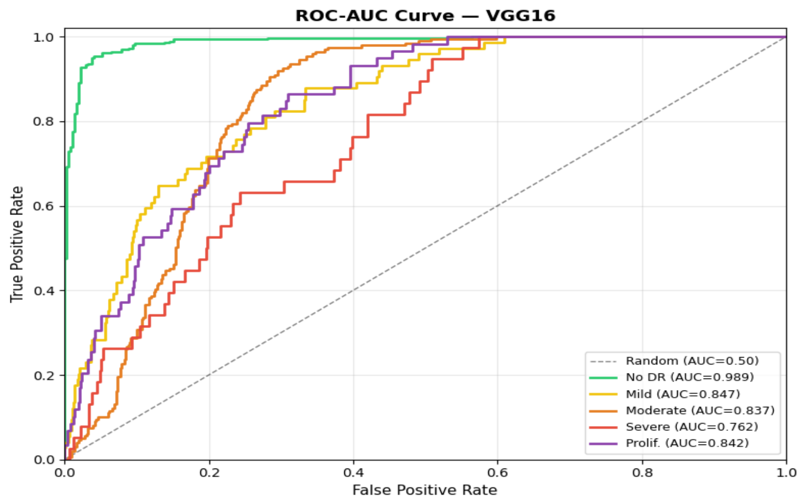

Figure 14. R0C-AUC of VGG16.

VGG16's ROC curves produced the lowest macro-averaged AUC of 0.8555 among all five models. Grade 1 and Grade 3 curves hug closest to the random diagonal, confirming the model's limited ability to discriminate these minority classes from others. The Grade 0 curve is the most favorable, reflecting the model's tendency to over-predict the majority class.

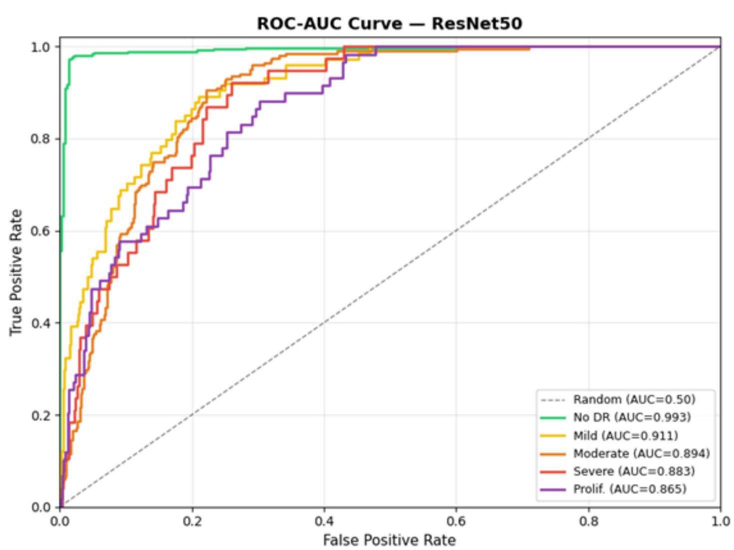

Figure 15. ROC-AUC of ResNet50.

ResNet50 achieved the highest macro-averaged AUC of 0.9090, representing a meaningful improvement over VGG16. The curves for Grade 0 and Grade 2 are near ideal, while Grade 1 and Grade 3 curves show substantial lift above the diagonal compared to VGG16. The area-based discriminative performance confirms ResNet50 as the strongest overall model in this experiment.

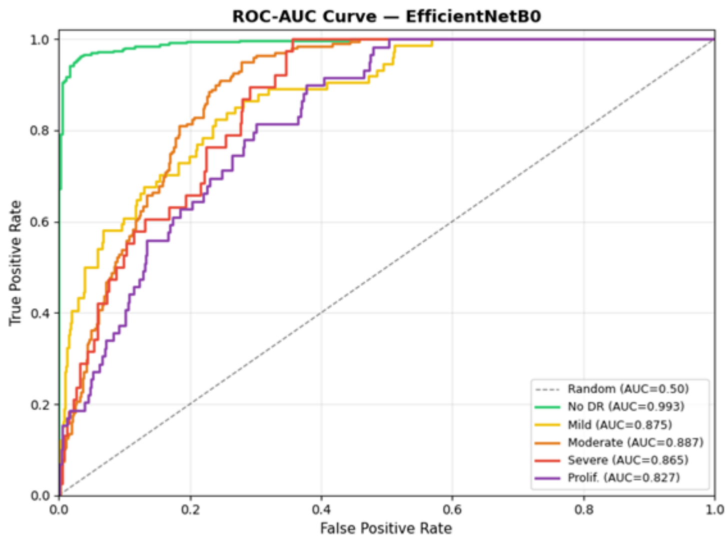

Figure 16. ROC-AUC of EfficientNetB0.

EfficientNetB0 achieved a macro-averaged AUC of 0.8892, the second lowest after VGG16. While the Grade 0 and Grade 2 curves are strong, the minority class curves reveal that the model's compact architecture and short training duration limited its ability to fully capture the probability distributions of rare classes. This is consistent with EfficientNetB0 requiring more epochs to realize its full potential.

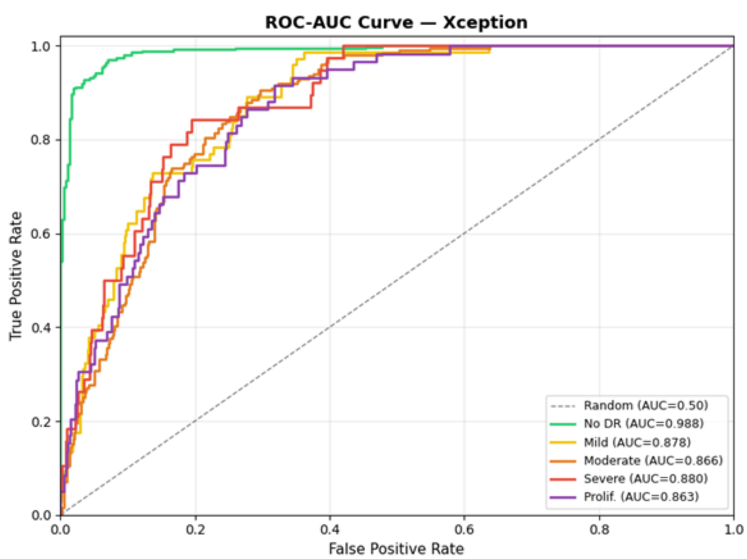

Figure 17. ROC-AUC of Xception.

Xception achieved an AUC of 0.8950, slightly above EfficientNetB0. The Grade 4 ROC curve showed particularly strong separation, consistent with Xception's ability to model global structural features like neovascularization. Grade 1 discrimination remained weaker, aligning with the confusion matrix analysis.

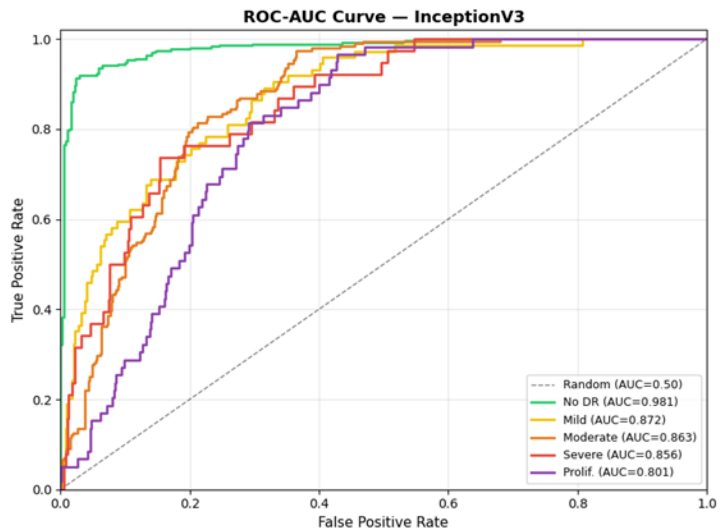

Figure 18. ROC-AUC of InceptionV3.

InceptionV3's macro-averaged AUC of 0.8746 placed it third overall. The multi-scale inception modules contributed to competitive Grade 4 discrimination, but the model underperformed ResNet50 and Xception on most other classes. The AUC profile suggests that given additional training epochs and fine-tuning of its upper convolutional blocks, InceptionV3 could potentially achieve higher discrimination.

4.5 Multi-Metric Comparison Charts

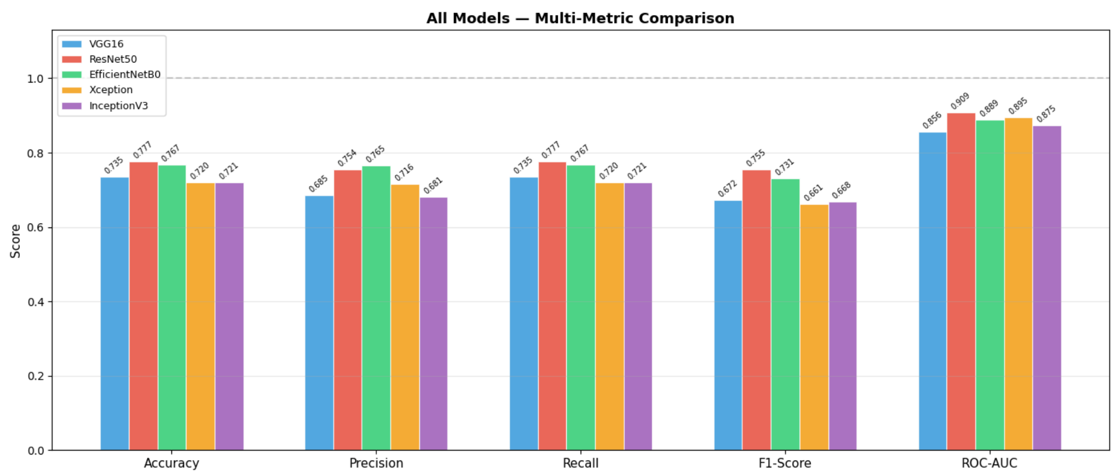

Figure 19. Grouped bar chart comparing all five models across all five evaluation metrics.

The grouped bar chart provides a unified cross-model comparison at a glance. ResNet50 leads on four of the five metrics, with EfficientNetB0 achieving the highest precision. VGG16 consistently occupies the bottom position. Notably, ROC-AUC scores are uniformly higher than F1-scores across all models, which reflects the fact that AUC measures ranking performance across all probability thresholds, whereas F1-score measures classification quality at the default 0.5 threshold where class imbalance has a more direct negative impact.

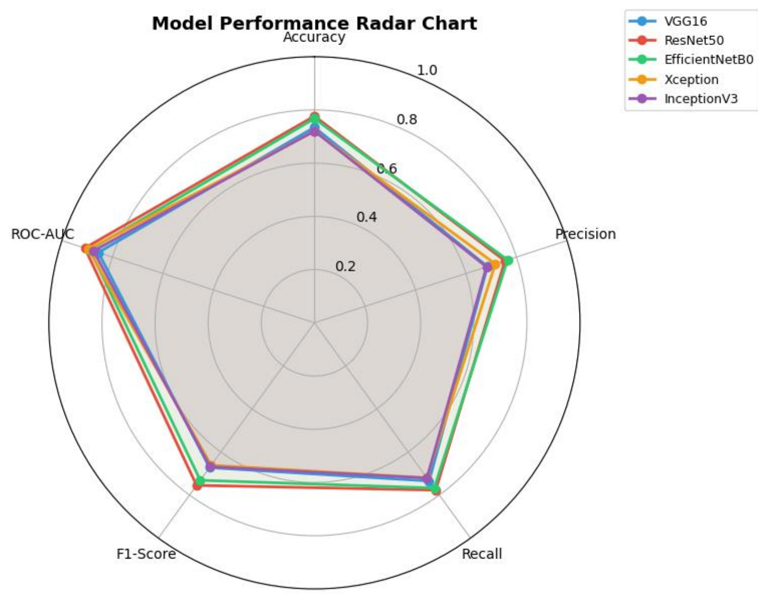

Figure 20. Radar chart showing the holistic performance polygon of each model.

The radar chart offers an intuitive view of each model's performance profile across all five dimensions simultaneously. ResNet50's polygon covers the largest total area, confirming its overall dominance. EfficientNetB0's polygon is slightly smaller but more symmetric, indicating consistent performance across all metrics. VGG16's polygon is the smallest and most irregular, underperforming especially on precision and F1-score. Xception and InceptionV3 show nearly overlapping polygons, suggesting similar capability profiles despite their different architectural designs.

5. Conclusion and Future Work

5.1 Key Findings

This project demonstrated that transfer learning is a highly effective strategy for automated diabetic retinopathy classification on the APTOS 2019 dataset. Among the five architectures evaluated, ResNet50 achieved the best overall performance with an F1-score of 0.7550 and a ROC-AUC of 0.9090, driven by its residual skip connections that ensure stable gradient flow and strong generalization on a relatively small dataset. EfficientNetB0 achieved the highest precision (0.7647), making it the most conservative predictor and the preferable choice in screening contexts where false positives carry high clinical cost.

VGG16, despite its historical significance, showed clear signs of over fitting due to its 138 million parameters. Xception and InceptionV3 performed comparably to each other and occupied a competitive mid-tier position. The results confirm that architectural innovations such as residual connections, compound scaling, and depth wise separable convolutions are more impactful for medical imaging tasks than raw model size or depth.

5.2 Impact and Limitations

The system developed here represents a proof-of-concept automated DR screening pipeline. Even at the accuracy levels achieved in this study, such a system could meaningfully reduce the burden on ophthalmologists in resource-constrained settings by triaging low-risk patients and prioritizing severe cases for specialist review.

Several limitations must be acknowledged. The class imbalance was not explicitly addressed through class weighting, focal loss, or oversampling techniques such as SMOTE, which likely contributed to the consistently low recall on Grades 1 and 3. The maximum training duration of 5 epochs was a computational constraint that may have disadvantaged deeper architectures like InceptionV3. The evaluation was performed on a single train-validation split, introducing variance that cross-validation would mitigate. No fine-tuning of the pertained backbone layers was performed.

5.3 Future Directions

Several promising directions exist for future improvement. Fine-tuning the top convolutional blocks of ResNet50 and EfficientNetB0 after initial feature extraction training could yield additional accuracy gains. Addressing class imbalance through class-weighted cross entropy loss or focal loss would likely improve minority class recall, which is the most clinically critical metric. Extending training to 20 or more epochs with a cosine annealing learning rate schedule would allow deeper models to converge more fully. Incorporating Grad CAM visualization would provide clinical interpretability by highlighting the retinal regions driving predictions. Finally, ensemble methods combining ResNet50 and EfficientNetB0 predictions could further improve robustness and calibration.

7. References

[1] Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. Proceedings of the 36th International Conference on Machine Learning (ICML). arXiv:1905.11946.

[2] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep Residual Learning for Image Recognition. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). arXiv:1512.03385.

[3] Simonyan, K., & Zisserman, A. (2015). Very Deep Convolutional Networks for Large-Scale Image Recognition. Proceedings of the International Conference on Learning Representations (ICLR). arXiv:1409.1556.

[4] Chollet, F. (2017). Xception: Deep Learning with Depthwise Separable Convolutions. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). arXiv:1610.02357.

[5] Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J., & Wojna, Z. (2016). Rethinking the Inception Architecture for Computer Vision. Proceedings of CVPR. arXiv:1512.00567.

[6] APTOS 2019 Blindness Detection Dataset. Kaggle Competition. https://www.kaggle.com/c/aptos2019-blindness-detection

[7] Gulshan, V., Peng, L., Coram, M., et al. (2016). Development and Validation of a Deep Learning Algorithm for Detection of Diabetic Retinopathy in Retinal Fundus Photographs. JAMA, 316(22), 2402-2410.

[8] Abadi, M., et al. (2016). TensorFlow: A System for Large-Scale Machine Learning. OSDI Conference Proceedings.

[9] Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825-2830.

[10] International Diabetes Federation. (2021). IDF Diabetes Atlas, 10th Edition. Brussels, Belgium.